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INFLUENCE OF THE PRIMARY JETS AND FUEL INJECTION ON THE AERODYNAMICS OF A PROTOTYPE ANNULAR GAS TURBINE COMBUSTOR SECTOR

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ABSTRACT

Transverse dilution jets are widely used in combustion systems. The current research provides a detailed study of the primary jets of a realistic annular combustion chamber sector. The combustor sector comprises an aerodynamic diffuser, inlet cowl, combustion dome, primary dilution jets, secondary dilution jets and cooling strips to provide convective cooling to the liner. The chamber contracts toward the end to fit the turbine nozzle ring. 2D PIV is employed at an atmospheric pressure drop of 4% (isothermal) to delineate the flow field characteristics. The laser is introduced to the sector through the exit flange. The interaction between the primary jets and the swirling flow as well as the sensitivity of the primary jets to perturbations is discussed. The perturbation study includes: effect of partially blocking the jets, one at a time, the effect of blocking the convective cooling holes, placed underneath the primary jets and shooting perpendicular to it. In addition, the effect of reducing the size of the primary jets as well as off-centering the primary jets is explained. Moreover, PIV is employed to study the flow field with and without fuel injection at four different fuel flow rates. The results show that the flow field is very sensitive to perturbations. The cooling air interacts with the primary jet and influences the flow field although the momentum ratio has a 100:1 order of magnitude. The results also show that the big primary jets dictate the flow field in the primary zone as well as the secondary zone. However, relatively smaller jets mainly influence the primary combustion zone because most of the jet is recirculated back to the CRZ. Also, the jet penetration is reduced with 25% and 11.5 % corresponding to a 77% and 62% reduction of the jet's area respectively. The study indicates the presence of a critical jet diameter beyond which the dilution jets have minimum impact on the secondary region. The jet off-centering shows significant effect on the flow field though it is in the order of 0.4 mm. The fuel injection is also shown to influence the flow field as well as the primary jets angle. High fuel flow rate is shown to have very strong impact on the flow field and thus results in a strong distortion of both the primary and secondary zones. The results provide useful methods to be used in the flow field structure control. Most of the effects shown are attributed to the difference in jet opposition. Hence, the results are applicable to reacting flow.

1 INTRODUCTION

Over the past decade, regulations on emissions increased drastically. New combustion concepts evolved to alleviate the emissions problem. Among the successful new concepts are the Lean Premixed (LP) [1-2] and Lean Direct Injection (LDI) [1-4] concepts. However, a series problem with new and conventional combustion concepts is the generation of thermo-acoustic instabilities (also known as combustion dynamics). It is a prominent problem in the case of LP combustors because of the loss of the damping mechanism that accompanies the diffusion process in non-premixed combustion.

Aerodynamics significantly affects stability, emissions and dynamics of Gas Turbine Combustors (GTC). Usually flame anchoring is achieved by applying swirl to the incoming air which results in the generation of a Central Recircualtion zone (CRZ). Great deal of research has been conducted on the CRZ structure and size (for example: ref [5-10]).

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Interaction between swirling flow and dilution jets has a strong impact on the flow field structure as well as dynamics and emissions. Primary dilution jets terminate the CRZ and complete the combustion process. It is a common practice to design the dilution jets to penetrate at least 40% of the primary zone width [11], [12]. Secondary dilution jets follow for trimming the exit temperature profile to match the turbine thermal loading requirements (also known as required pattern factor). Richards and Samuelsen reported that the location of the transverse jet is critical to the combustion process. They stated that a placement of the transverse primary jets downstream of the swirler exit with half to one duct diameter will results in optimum penetration [13-14].

The current study is conducted on isothermal flow. The experimental setup is very challenging and reacting flow experiments are very expensive, if possible. The literature (for example references [15-21]) shows that the differences between isothermal and reacting flow are located in the primary zone close to the swirler exit. However, the secondary and dilution zones have several similarities. The turbulence levels in these regions are very similar. Also, in both modes the dilution jets dictate the combustor exit velocity profiles.

This paper presents a first study of the transverse jets and their influence on the flow field structure through a set of experiments conducted using Particle Image Velocimetry (PIV) on the flow field of a single cup fuel rich dome sector (also referred to as Single Annular Combustor or SAC). The flow field is studied under partial blocking of the primary jets as well as off centering the primary jets. In addition, the interaction of the convective cooling film with the primary jets is investigated. Also, the effect of the primary jet size on the flow field structure is presented. Finally, the effect of fuel injection on the flow field is discussed. The results provide useful methods to be used in the flow field structure control. Most of the effects shown are attributed to the difference in jet opposition. Hence, the results are applicable to reacting flow.

2 NOMENCLATURE

Va	=	Axial velocity component
Vr	=	Radial velocity component
V	=	Total velocity defined as $V = \sqrt{(Va)^2 + (Vr)^2}$
RMSR	=	Root mean square value of the radial velocity
RMST	=	Root mean square value of the tangential velocity
R	=	Flare exit radius
CCD	=	Charge Coupled Device
CFD	=	Computational Fluid Dynamics
SAC	=	Single Annular Combustor
pphr	=	flow rate in Pounds per hour
PIV	=	Particle Image Velocimetry
LDV	=	Laser Doppler Velocimetry
LP	=	Lean Premixed Prevaporized
O2	=	Oxygen
N2	=	Nitrogen
LDI	=	Lean Direct Injection
GTC	=	Gas Turbine Combustor
CRZ	=	Central Recirculation Zone
DPSS	=	Diode Pumped Solid State
X, Y, Z	=	Cartesian coordinate system with the origin defined at the flare center

3 EXPERIMENTAL SETUP AND TEST CONDITIONS:

3.1 Swirl cup and test section:

The counter rotating coaxial radial swirl cup arrangement used in this study is illustrated in Figure 1. The inner swirler is connected to a venturi tube. The area ratio of the secondary to the primary swirler is greater than 1. The secondary swirler is connected to a conical flare. The effective area of the swirler cup is 0.62 in^2. The swirl cup has a splash plate and a combustion dome attached to the exit as depicted in Figure 2. The dome expansion angle is not symmetric. There is a difference of 9 degrees in the expansion angle of both sides of the dome. The SAC sector used is similar, in construction, to the one shown in Figure 3. The sector contains two side quartz windows, primary and secondary dilution jets, eight film cooling strips on each side, combustion dome with cooling slots, inlet cowl and inlet diffuser.

A stainless steel probe is used to study the air/fuel mixing. The probe couldn't be inserted from the exit flange because of the contraction that reduced the cross section significantly toward the SAC exit. As a result, the sampling probe is inserted through the side wall. However, to enable scanning the flow field in the X-Z plane, the side plate is designed as shown in Figure 4.

The side plate is composed of aluminum strips that are assembled together using two aluminum holders. The sampling probe is then inserted through a hole in a relatively longer movable aluminum strip as depicted in Figure 4. The movable strip is attached to the computer-controlled traverse to enable scanning of the flow field in the X-direction. In other words, we had to scan a complete line in the X-direction and then switch the movable strip with the upper or lower strip to be able to scan the flow field at a different Z.

3.2 Experimental facility:

A schematic of the facility utilized is presented in Figure 5. The SAC sector is mounted on a flange above an 8 inch PVC pipe used as a manifold. A honeycomb inside the manifold provides uniform flow to the SAC diffuser inlet. Pressure sensor and thermocouples are mounted inside the manifold for monitoring of pressure and temperature at the SAC sector inlet. Air enters the manifold through a 2 inch pipe which can provide pressurized air up to 2lb/s at 100Psi. An atomizer generating olive oil seeding particles (1~5 μm) is connected to the manifold though a seeding port. A blower and an exhaust system are mounted above the SAC sector to collect aerosols from the combustor.

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3.3 Diagnostics:

The flow field velocity measurements are conducted using PIV. The PIV system is provided by Lavision Inc. The system includes two 1376 x 1040 pixels 12 bit Lavision imager intense CCD cameras and two 120 mJ 15 Hz pulse Nd-Yag lasers which are synchronized with CCD cameras. A mirror is used to get the laser beam to the sheet optics and provide the laser sheet at the correct measurement plane as illustrated in Figure 6. Both cameras are placed on the top of one another to extend the measurement volume (see Figure 7). Both cameras are placed perpendicular to the laser sheet. The set up is a real challenge since the exit port of the SAC sector has reduced width to enable it to fit to the turbine nozzle ring. Figures 6 and 7 illustrate how complex is the PIV setup.

Fuel air mixing is studied using a sampling probe connected to an Oxygen (O2) paramagnetic analyzer for O2 measurement. 2D PIV is used to delineate the flow field characteristics with and without fuel injection.

3.4 Test conditions, data acquisition, measurement grid and uncertainty:

The measurements coordinate system is defined such that the axial direction of the swirler is designated as the Z direction with the zero set at the flare exit. The X direction perpendicular to the Z and in the dilution air jet plane. The origin of this coordinate system is located at the center of the swirler exit. The coordinate system is shown in Figure 2. Experiments are carried out for isothermal flow condition under ambient pressure and temperature of 70±1ºF.

PIV velocity measurements are conducted in the X-Z plane passing by the SAC center (Y=0) at a pressure drop of 4.3%. The corresponding air flow rate is approximately 1000 pphr. The average flow field is obtained by averaging of 200 PIV instantaneous images. An interrogation area size of 32x32 pixels with a window overlap of 50% is used for the image processing algorithm.

The air fuel mixing measurements are undertaken at a fuel flow rate of 25 pphr. The SAC sector is scanned with a resolution of 3mm in the X direction. Measurements are conducted at Z/R= 0.2, 0.58, 0.96, 1.34, 1.72, 2.49 and 3.25 respectively.

PIV is also employed to study the effect of fuel injection on the flow field. The measurements are conducted at fuel flow rates of 25, 35, 45 and 55 pphr.

The authors used 50, 75, 100, 150 and 200 images in processing the PIV images. The best resolution and minimum change in average are noticed as the number of images approach 150 images. The error in average is known to be: (standard deviation/(number of images)^0.5. Consequently, the authors increased the number of images to 200 in order to bring the error to a minimum while maintaining reasonable time in acquiring and processing the data.

The laser sheet is focused in the primary zone and close to the primary jets (region of high gradients). As reported in the literature, as long as a minimum of 10 seeding particles occur within the interrogation area then the removal of the Spurious vectors using the deviation from the local median (and replacing either by vectors corresponding to the 2nd, 3rd or 4th highest correlation peak) should result in a typical uncertainty of 5-10% [22].

4 RESULTS AND DISCUSSIONS

4.1 The effect of partially blocking the primary dilution jets strip:

The SAC sector has two strips with holes that provide the primary dilution jets. There are two strips that oppose one another. Each strip is blocked, one at a time, and PIV is conducted on the SAC sector to study how the blockage process can be used to control the flow field structure. The blocking process is illustrated in Figure 8.

The velocity contours and streamlines with and without blockage are illustrated in Figure 9. Figure 9 (a) shows the flow field that results when the left primary jet is partially blocked. Figure 9 (b) shows the flow field that corresponds to all jets fully open. Figure 9 (c) shows the flow field corresponding to the partially blocked right primary dilution jet strip.

With the partially blocked left jet, we notice that the left jet becomes weak relative to the right jet and hence, the impingement point is shifted from the SAC centerline to the left half of the combustor as depicted in Figure 9 (a). This corresponds to a reduction of 9.2% in the jet penetration. A strong asymmetry occurs in the SAC primary zone. As the right primary jet penetrates more in the primary zone it controls and dictates the structure of the SAC primary zone as illustrated in Figure 9 (a). It is important to note that as the left dilution jet becomes weaker, the pressure applied from the swirling flow results in a larger curvature of the left jet trajectory. In other words, with the partially blocked jet, the jet angle, with the horizontal, increases by 6.5°. However, we notice that the size of the CRZ is roughly the same with or without blockage.

It is interesting to also note that the effect of the partially blocked left dilution jet extends to the secondary region or to the post recirculation zone as shown in Figure 9 (a). This becomes obvious when we compare Figure 9 (a) and 9 (b). Without blockage, the swirling flow mixes with the primary jets and is almost shooting vertical in the secondary region in the vicinity of the SAC Sector centerline. However, with the partially blocked left jet, the flow in the secondary region tilts toward the left hand side of the SAC sector with an angle of -7 ° (relative to the original position). The –ve signs means in anticlockwise direction.

All of the aforementioned effects might have a strong impact on the emissions, stability and dynamics of the SAC sector. However, reacting flow experiments are needed to quantify and study the effect in greater details.

The effect of the partially blocked right dilution jet on the flow field could be explained in a similar manner and therefore, for reasons of brevity, we don't reiterate the same discussion here. However, we report a quantification of the effect on the flow field: the right primary jet angle increases with 3.5°, the left jet penetration increases by 7.5% and the jet impingement

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region tilts with an angle of +20 ° (relative to the original position). +ve refers to rotation in the clockwise direction.

4.2 The effect of blocking the cooling holes underneath the primary dilution jets:

The primary dilution jets are provided through holes drilled on the liner cooling strip. The cooling strip is a strip that is cooled by convective cooling. Holes with a diameter in the order of magnitude of 1 mm provide the cooling air necessary to protect the liner from melting. The diameter of the hole providing the primary jet is around 15 times the hole providing the cooling jet. The concern is raised about what kind of interaction that takes place between the primary jets and the cooling jet if any. This is of concern because the cooling jet is shooting perpendicular to the primary dilution jet at the point where it leaves the hole as demonstrated in Figure 10. Such a question is of immense importance especially in the combustor design process as well as flow modeling techniques. As a matter of fact the couple of holes placed right underneath the jets are useless from the point of view of liner cooling (see Figure 10). Simply there is no metal at this location due to the presence of the dilution hole. Then, the major importance of the holes underneath the jets lies in the possibility of them being used to control the flow field structure as we will show shortly.

Figure 11 shows the velocity contours and streamlines with and without blockage of the cooling strip underneath the primary dilution jets. Figure 11 (a) shows the case with a blocked left cooling strip, while Figure 11 (b) shows the flow field without blockage and Figure 11 (c) shows the flow field with a blocked right cooling strip.

Under normal running conditions and without blockage, the cooling air jet interacts with the primary jet and pushes against it resulting in a slight increase of the primary jet angle along the horizontal. As the left cooling jet is blocked, we notice a slight decrease in the jet angle and a corresponding change in the flow field in the primary as well as the secondary zone as depicted in Figure 11 (a). Without blockage, both primary jets impinge and a portion of the resulting flow is recirculated back in the CRZ while the other portion shoots up in the secondary region. With blockages, in the vicinity of the centerline, the CRZ centerline tilts with an angle of +8° relative to the original direction. The jet impingement region angle increases with + 6°. The effect of blocking the right cooling strip can be explained in a similar manner. The associated change is manifested in an increase in the tilt of CRZ centerline with an angle of +14.5° relative to the original direction. Also, the jet impingement region angle increases with an angle of -8.4°. The momentum of the primary jet is more than 100 times the momentum of the cooling jet. However, it still affects the flow field as discussed previously.

4.3 The effect of off-centering the primary jets on the flow field.

In the previous sections, it has been shown that the primary jets have a strong influence on the flow field structure. The aim of the current section is to study the effect of the slight primary jets off centering on the flow field. This study is useful for two reasons: First, it provides a method for flow control. Second, as engines age, such asymmetry in the geometry is inevitable due to factors like thermal stresses and liner oxidation.

The off-centering is achieved by initially reducing the primary jet hole diameter from 0.65 to 0.4 and then we off center the jets asymmetrically such that the inner hole is placed off center by 1.4 mm and the outer hole is placed off center by 1.0 mm as illustrated in Figure 12.

The off centering results are presented in Figure 13. The figure shows the total velocity contours with the jets centered and off centered. With the jets centered, we notice that the flow field is fairly symmetric. However, with the jets off centered, the right jet is slightly larger in diameter than the left jet in the measurements plane (recall the off centering is not symmetric). It is very interesting to see that the higher strength of the right jet is pronounced in more jet penetration and more influence on the flow field as depicted in Figure 13. The right jet penetration increases with 11% while the left jet penetration decreases by 8.5%. There is not much opposition from the left jet, hence the flow field in the primary zone is dictated by the right jet. This is manifested an increases in the CRZ centerline tilt with an angle of +27.5° relative to the original location. Also, the Jet impingement region tilts with an angle of +21.5° relative to the original direction.

4.4 The effect of primary jet size on the flow field

The original size of the primary jet holes were 0.65 inch. We conduct the measurements using jets with diameters of 0.4 and 0.31 inch to study the effect of the primary jet size on the SAC flow field. The jet area reduction that associates the diameter reduction is 62% and 77% respectively. The total velocity contours and streamlines for all three cases are shown in Figure 14. As the jet diameter decreases, the momentum decreases significantly. Consequently, the jet penetration decreases by 11.5 and 25 % corresponding to the 62 and 77% reduction in the jet area.

For the big jets, a small percentage of the jet is recirculated back to the CRZ while a larger percentage of the jets mix in the secondary region and dictate the flow field structure therein. The SAC sector cross sectional area starts to decrease toward the outlet after approximately 4R. However, this contraction in the area doesn't seem to affect the flow field in the vicinity of the combustor centerline in the secondary region (for the big jet's). We can see that the flow resulting from the jet impinge is almost shooting up vertical in the case of the big jet.

As the jet size is reduced, we notice that the unrecirculated portion of the primary jets decreases significantly and so does the influence on the primary jets on the secondary region. The 62% reduction in area (0.4 inch jets) results in a tilt in the CRZ with +7° and a tilt in the jet impingement region with +6°. On the other hand, the 77% reduction in the jet area (0.31 jets) results in a tilt in the CRZ with +16.5° and a tilt in the jet impingement region with +16° as illustrated in Figure 14.

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For the 0.31 diameter jets, the tilting of the flow field to the right hand side appears to be the effect of the chamber contraction which now has more influence on the flow field in the secondary region than the primary.

These results emphasize the importance of the careful selection of the dilution jet's size. Not only is the location of the jet is of high importance, but also the size has significant effects on the flow structure and behavior in the primary as well as the secondary zone. It appears that for every configuration, a critical diameter of the dilution jets exists. By critical we mean the diameter below which the influence of the primary jets will be restricted to the primary zone.

4.5 The effect of fuel injection on the flow field

The profiles of fuel to air ratio by mass (F/A) at different downstream distances are presented in Figure 15. The measurements are done at 25pphr (Global F/A is 0.025).

The fuel concentration profile close to the flare exit (Z/R=0.2) shows sharp double humped profile as presented in Figure 15. At the same axial distance (Z/R=0.2), the measurements extend on the X direction from -1.2R to 1.2R (physical domain boundaries). In other words, beyond ±1.2R the dome surface is present. We also observe that at this axial location, the minimum F/A ratio is 0.05 (twice the global ratio). The concentration peaks and reach 0.12 (5 times the global ratio) at a radial distance of around 0.7R indicating the attainment of maximum fuel concentration. As we proceed downstream to an axial distance of 0.581R, the fuel spreads over the cross-section and peaks at a radial distance of ±1.2R as demonstrated in Figure 12. At the same location, Z=0.581R, the minimum F/A is still 0.05. The maximum F/A is around 0.09 (3.5 times of the global ratio).

The F/A profiles flatten at a downstream distance of 0.962R and show weak peaks at radial distances of -1.8R and 2.0R respectively. This asymmetry may be attributed to the asymmetry of the combustion dome expansion angle. We notice that the profiles are similar at distances of 0.96, 1.34 and 1.72 as shown in Figure 15. This indicates that most of the mixing takes place in a downstream distance of around 1.0R (similar conclusions are deduced from the F/A mixing profiles at fuel flow rate of 55 pphr, not shown here). However, as the dilution jets are encountered (Z/R=2.49 and 3.25), the F/A profiles drops significantly as tremendous amount of air is admitted through the holes.

Figure 16 shows the contours of the local equivalence ratio (ϕ_l) normalized by the global equivalence ratio (ϕ_g) imposed on the flow field velocity vectors. The black contours represent the edge of the CRZ and the primary jets wake regions respectively. This figure is of prime importance as it helps determine the stability characteristic of the combustor [23]. We notice that the fuel spreads in the vicinity of the edge of the CRZ with an angle of 45° (with the horizontal). The authors expect that a smaller spreading angle of 50 to 60 ° will result in higher concentration of the fuel inside the CRZ and since the velocities are lower, a highly sustained flame is highly probable. This might result in damping the oscillations observed during the reacting flow experiments [24].

It is important to study the impact of the fuel injection on the aerodynamics of the combustor. In the previous sections, we discussed the sensitivity of the primary jets to perturbations. We drew the general conclusion that the flow field is sensitive especially when it comes to perturbations that directly affect the primary jets. It is critical to keep in mind that this might not be a universal behavior because the size of the jets is relatively big in the combustor under study. This is illustrated in the study we conducted on the effect of the jet size on the flow field.

Figure 17 shows the velocity vectors and streamlines for the primary region with and without fuel injection. The fuel flow rate is 25 pphr. We observe that there is a change in the flow field structure with fuel injection. The total air flow rate at pressure drop of 4.3% is approximately 1000 pphr. Usually Fuel Rich Dome Combustors are designed such that 30% of the total air flow rate flows through the combustion dome (roughly 350 pphr). If the fuel flow rate is 25 pphr then the percentage of the fuel flow rate represents around 7% of the air flow rate.

To be able to provide a better explanation of the effect of fuel injection on the flow field, we construct a plot of the difference in velocity magnitudes with and without fuel injection. Figure 18 shows the value of V (with injection) -V without injection. We notice that the velocity increases in the swirling jet region because most of the fuel is present in this region as depicted in Figure 15. The velocity decreases in the vicinity of the primary jets upstream region and increases in the vicinity of the primary jets downstream region. This is due to the change in the primary jets angle (-3° relative to original location). It is clear from Figures 17, 18 and 19 that the jet angle increases continuously with fuel injection.

Figure 19 shows the velocity vectors and streamlines with and without fuel injection at fuel flow rate of 55 pphr. The edge of the CRZ and the primary jets are regions of high turbulence activities. Hence, once instabilities are generated, they usually occur at such regions. Indeed, such instabilities were observed while conducting reacting flow experiments on the same SAC sector [24]. The low frequency instabilities were attributed to the swirling flow in the primary region. The authors wanted to study the effect of fuel injection on the size of the CRZ because the CRZ is the source of ignition, any parameter that affects the CRZ (especially the upstream portion) may be significantly contributing to the generation of instabilities. The primary jets wake regions should be thought of in a similar fashion. In other words, the primary jet edges are zones of fuel reaction, if enough fuel is provided. This is evident from the reacting flow experiments that were conducted on the same SAC sector [24].

Figure 20 shows the size of the CRZ as well as the jet wake regions at different fuel flow rates. We can see that the size of the CRZ is approximately constant with a slight variation in the upstream region close to the flare exit. However, significant reduction of the jets wake regions takes place with fuel injection as presented in Figure 20. It also seems that the left primary jet wake region change continuously in size with the increase in the fuel flow rate. It is worth mentioning that a very

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similar behavior was observed in reacting flow experiments. Also, this behavior of the left jet wake region provides one possible explanation for the generation of instabilities that occurred in the primary jet regions at high power conditions. Thermo-acoustic instabilities studies on this SAC sector are discussed elsewhere [24].

5 CONCLUSIONS

Complex interaction takes place between the primary dilution jets and the swirling flow in GTC. The diameter of the jets is a critical parameter that dictates the degree of influence that the primary jets impose on the flow structure. Larger jets appear to influence both the primary and secondary regions. The influence of the smaller jets is limited to the primary region. The process appears to be related to, mainly, the portion of the jet recirculated back to the CRZ as well as the jet penetration.

Introducing perturbations with high amplitude (like partially blocking the jets or off-centering the jets) strongly influences the flow field structure mainly because of the reduction of the jet opposition as well as the change in the jet angle. Perturbations of small magnitude (like blocking the cooling jets underneath the primary jets) still affect the flow field, but only slightly.

Transverse jets contribute to the generation of instabilities (at high power) because the fuel flow influences the jet wake region and the heat release that occurs at this location. The effect is more pronounced at high fuel flow rate.

The results presented in the current research provide the industry as well as researchers in the field of GT combustion with the following:

- Methods to effectively control the flow field structure.
- Factors to be considered during the design process.
- Expected characteristics with combustor wear and aging.

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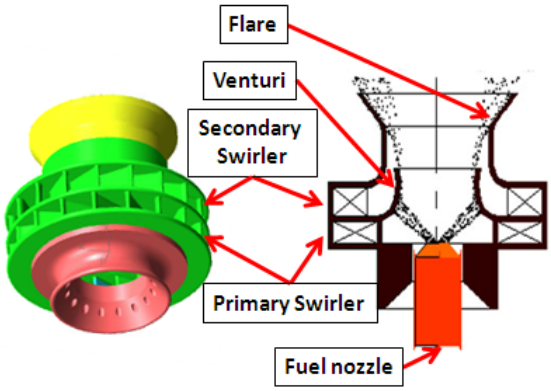

Figure 1. Typical swirl cup arrangement with counter rotating radial swirlers.

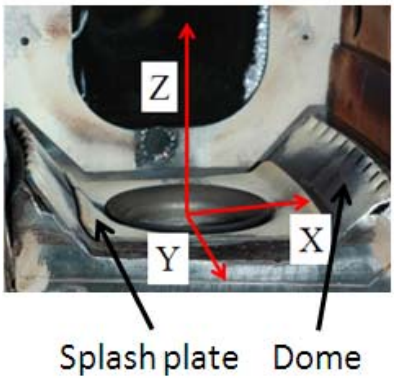

Figure 2. Swirl cup exit configuration and coordinate system illustration.

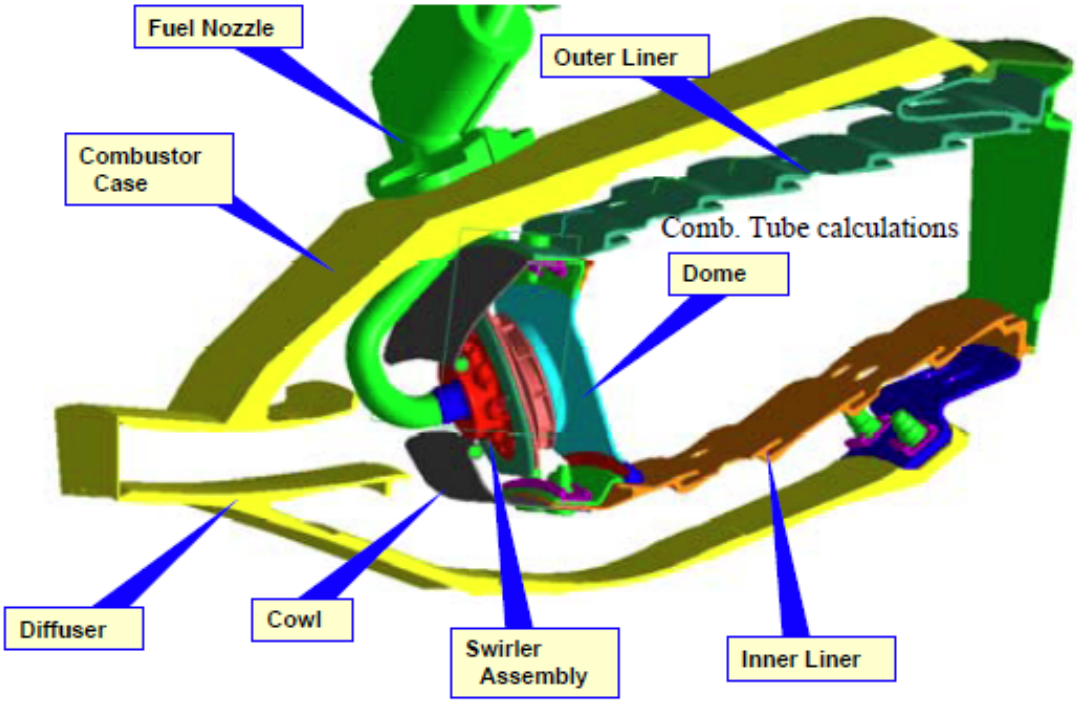

Figure 3. Typical SAC Sector.

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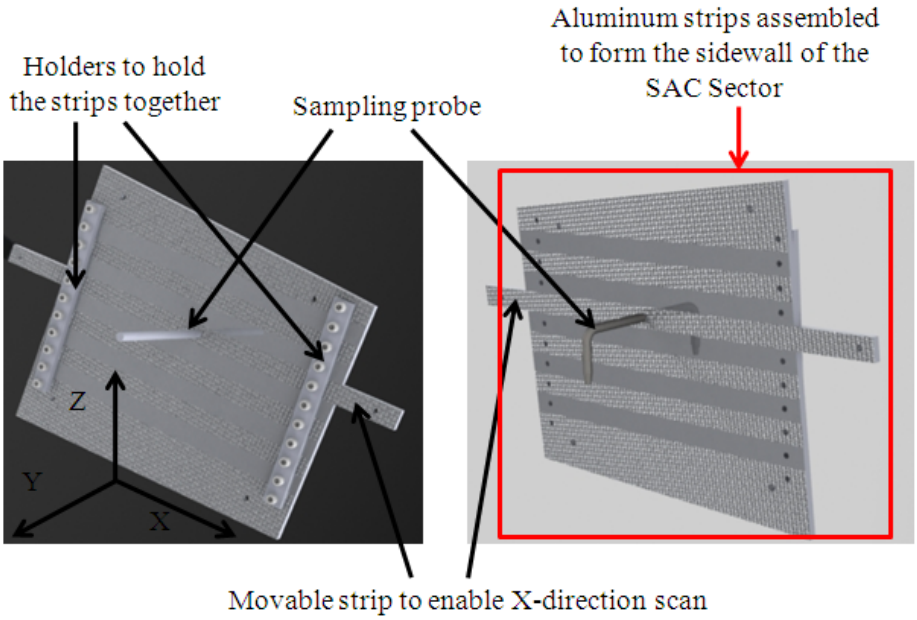

Figure 4. Side plate designed to enable insertion of the sampling probe and scanning of the SAC flow field.

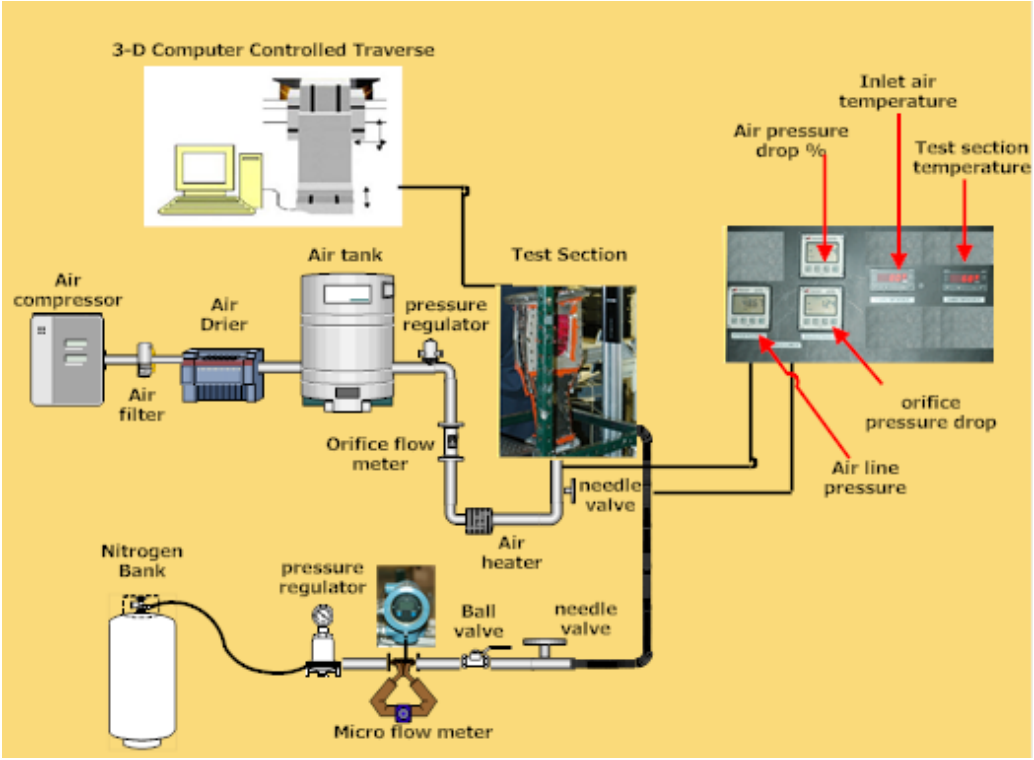

Figure 5. Schematic of the experimental facility.

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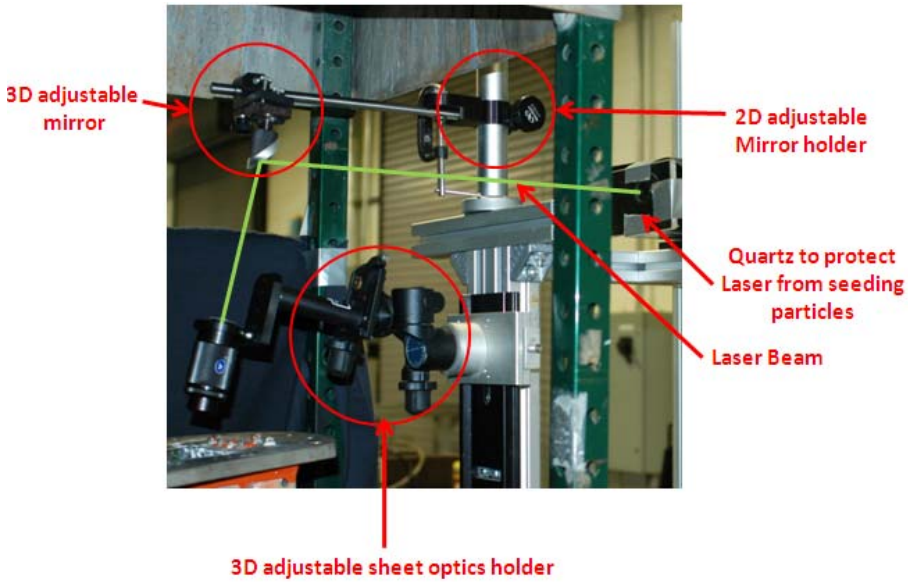

Figure 6. Optics arrangement to get the laser sheet inside the SAC sector

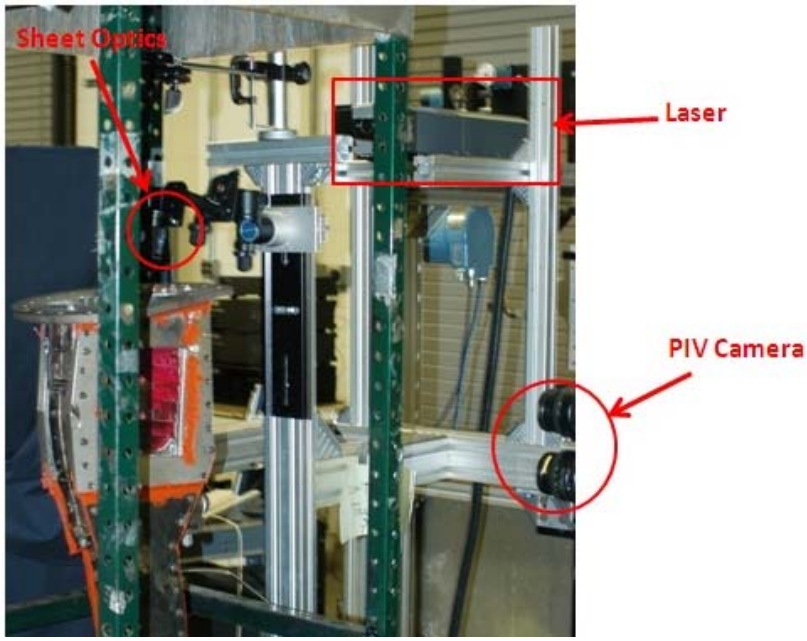

Figure 7. PIV setup on the rig.

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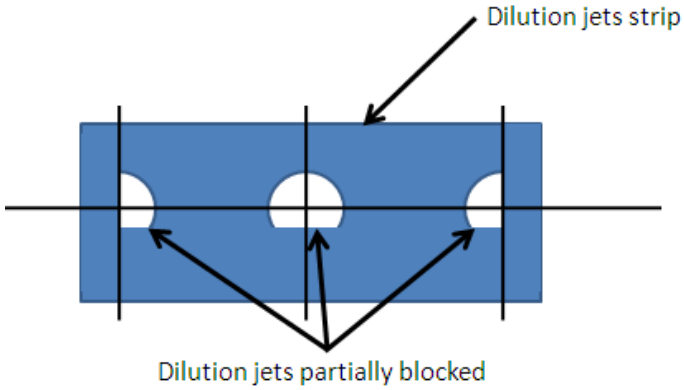

Figure 8. Illustration of the primary jets partial blocking.

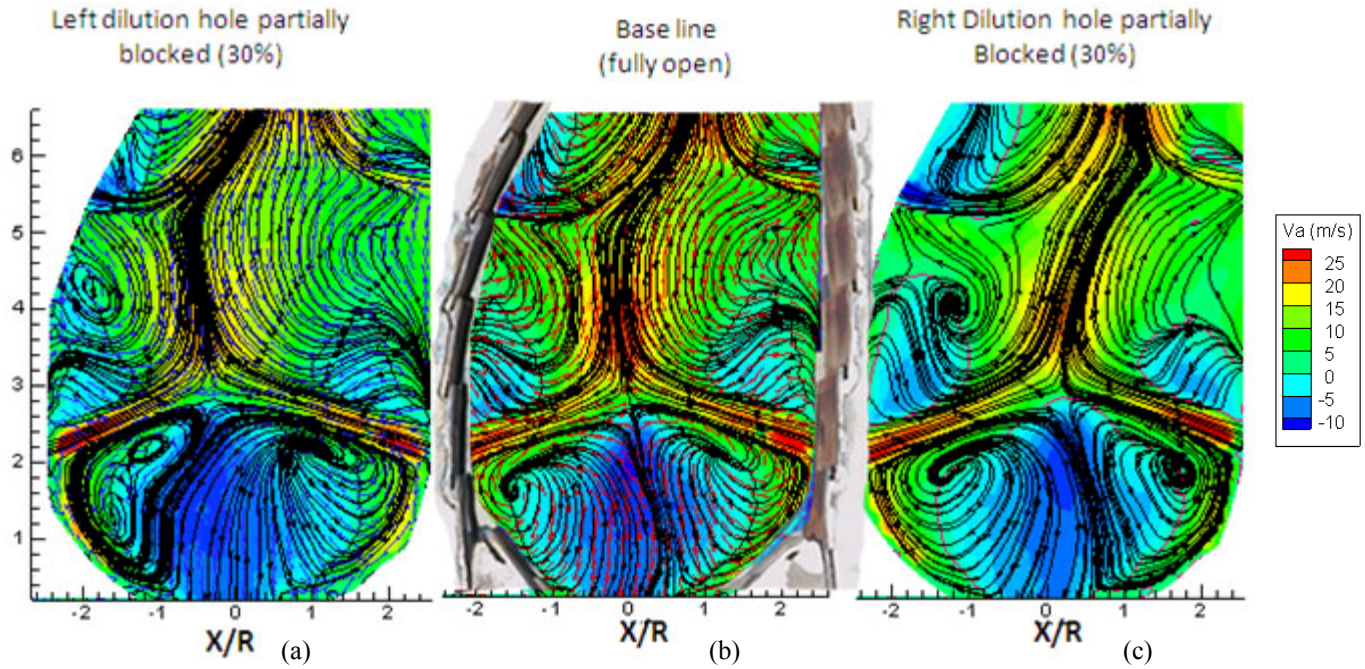

Figure 9. The effect of partially blocking the primary dilution jets on the flow field.

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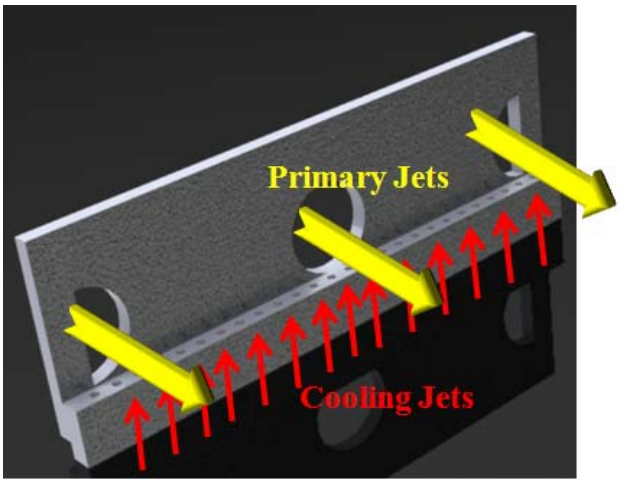

Figure 10. Illustration of primary jets and cooling jets interaction

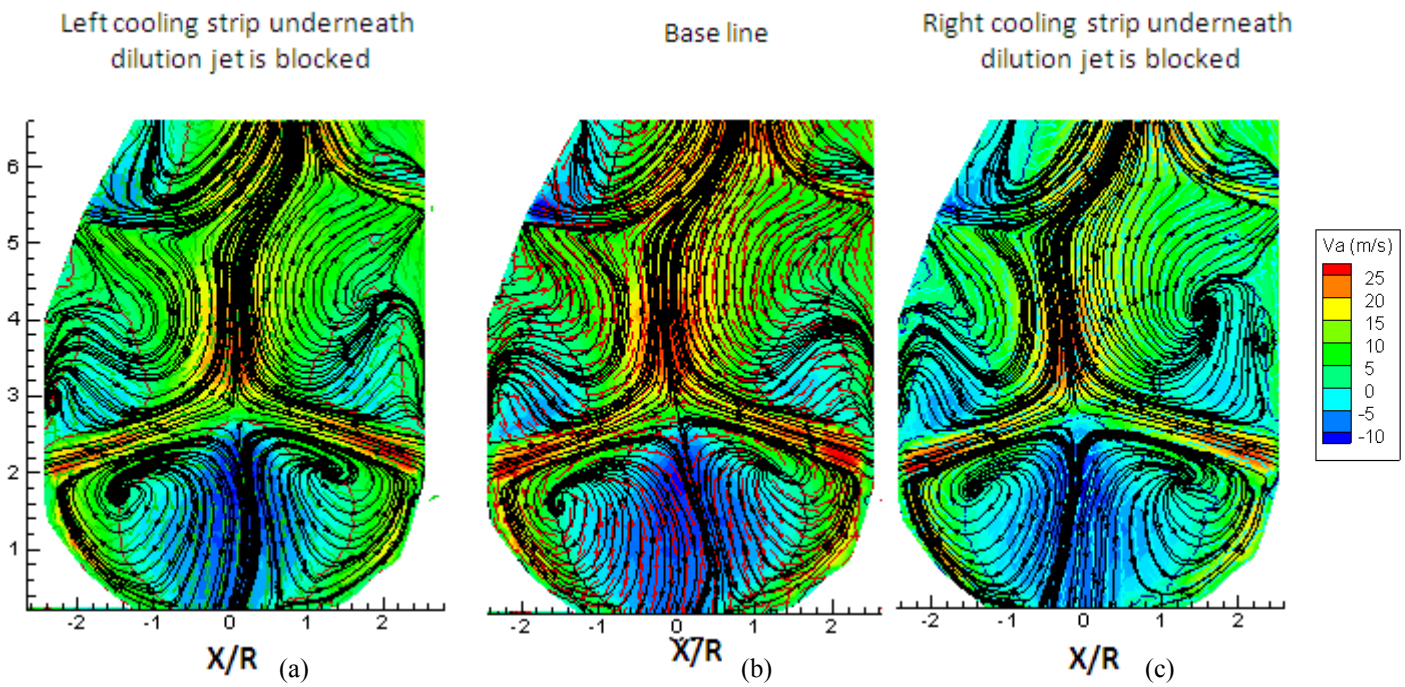

Figure 11. The effect of blocking the cooling strip underneath the primary jets.

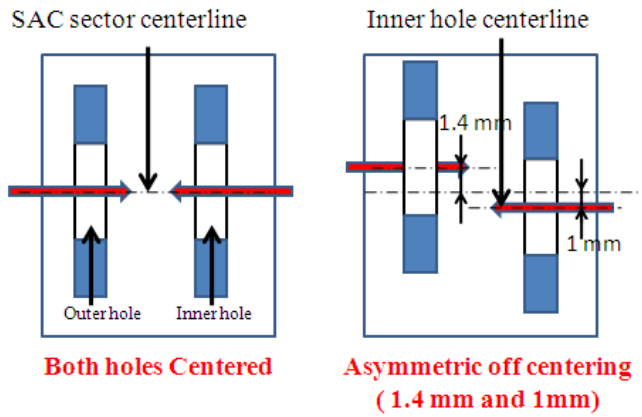

Figure 12. Illustration of primary jet holes off centering.

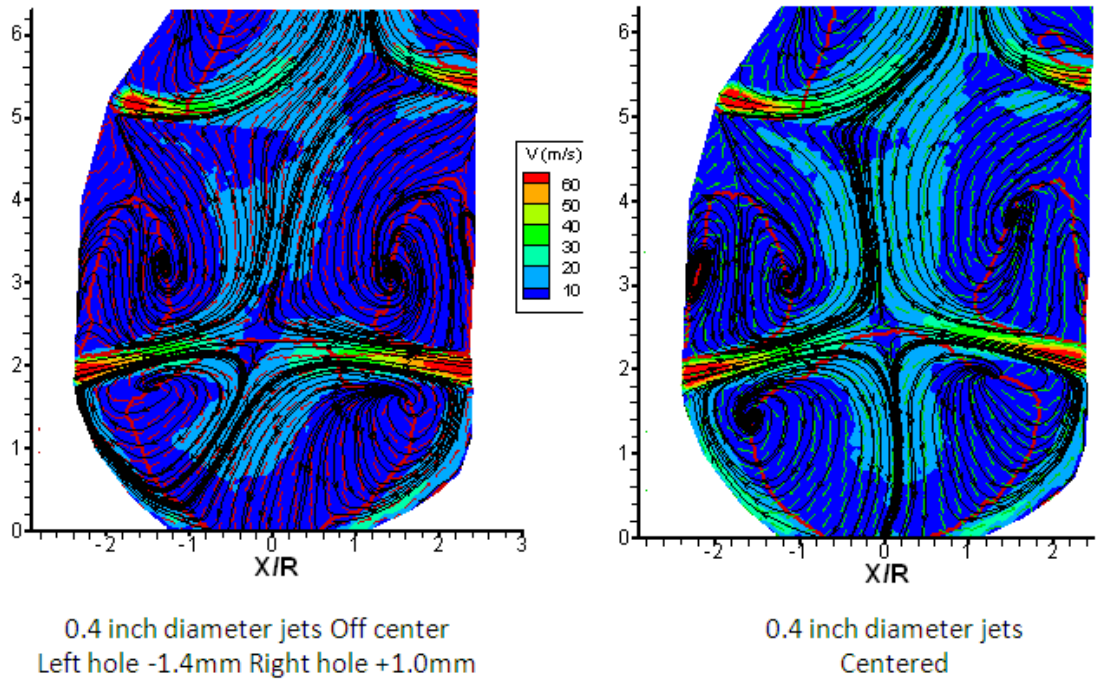

Figure 13. The effect of off-centering the primary jets on the SAC flow field

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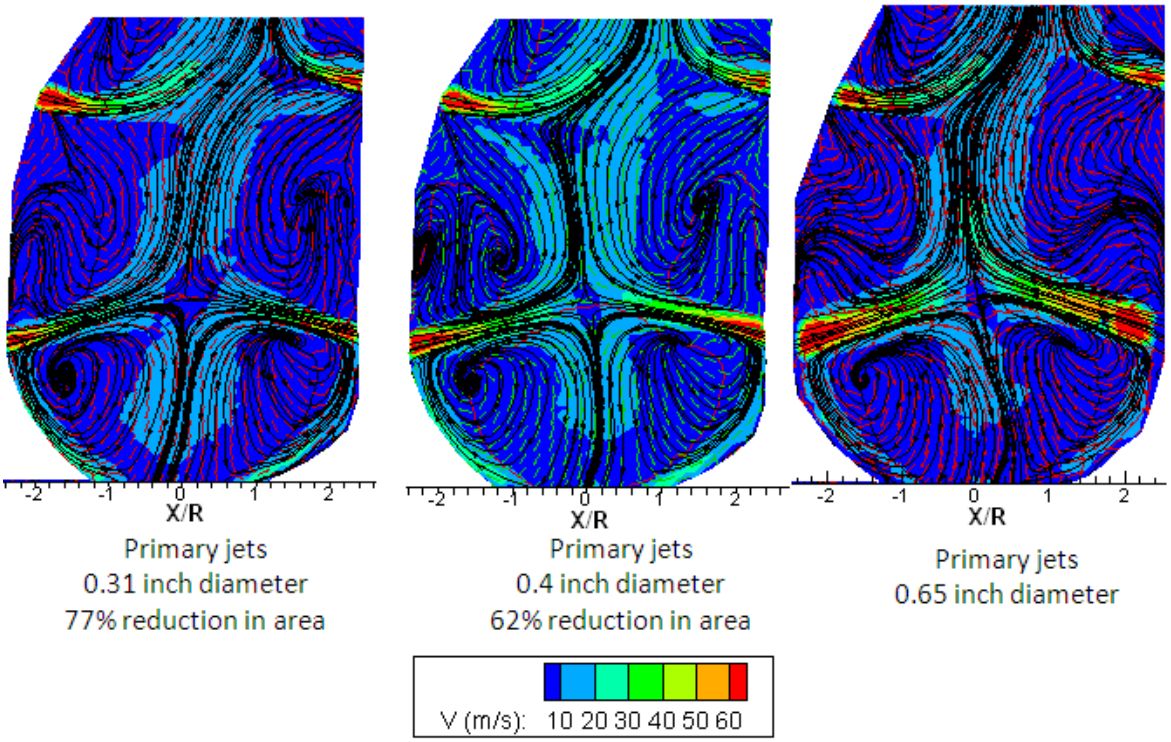

Figure 14. The effect of primary jet size on the flow field.

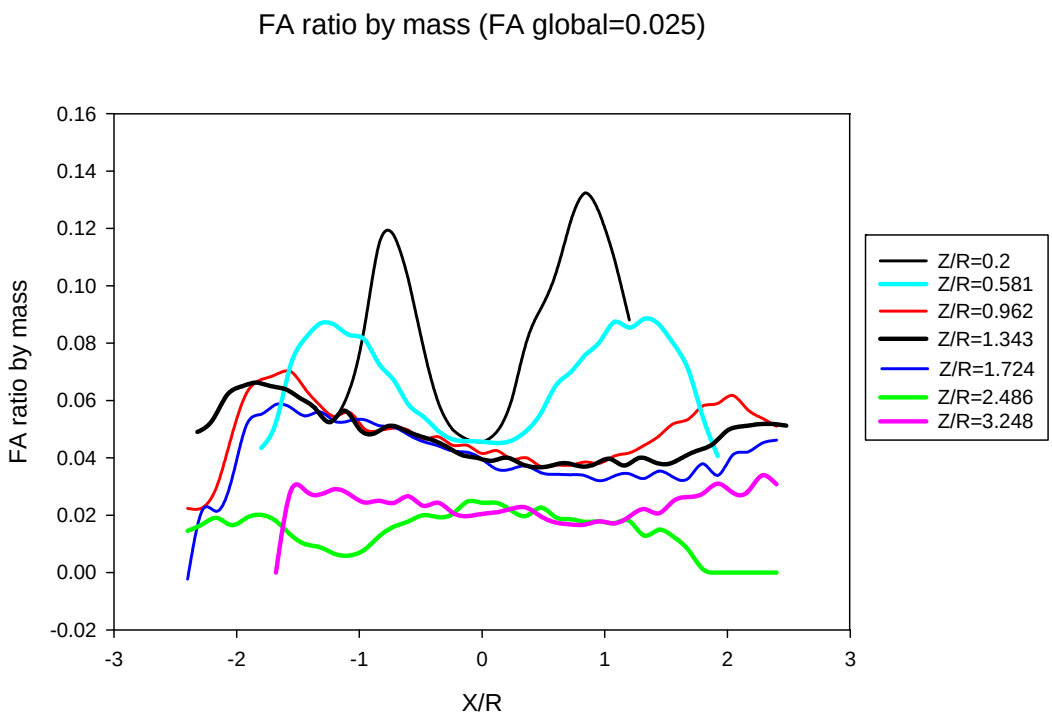

Figure 15. Profiles of F/A ratio by mass at fuel flow rate of 25pphr at different axial stations.

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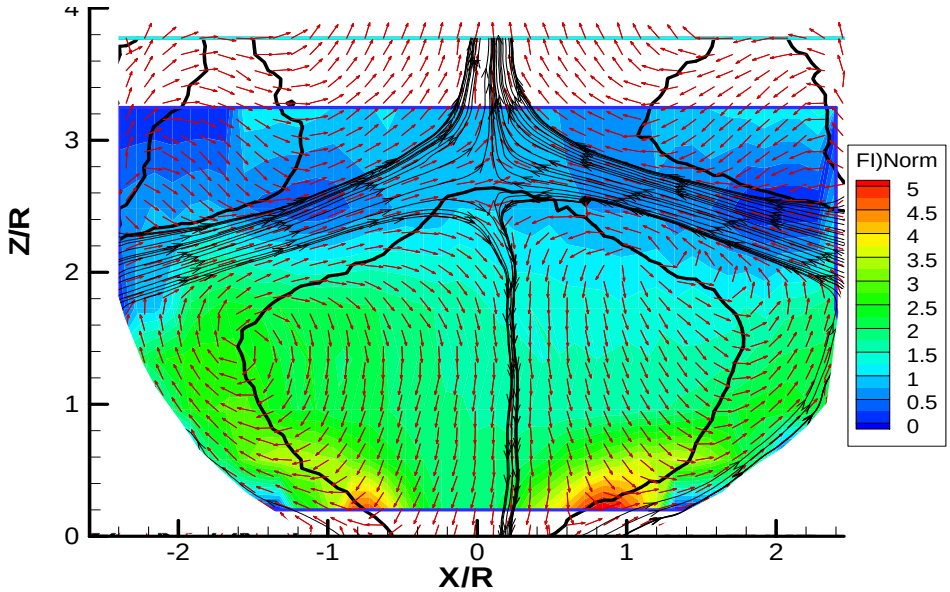

Figure 16. Normalized equivalence ratio contours imposed on velocity vectors at fuel flow rate of 25 pphr.

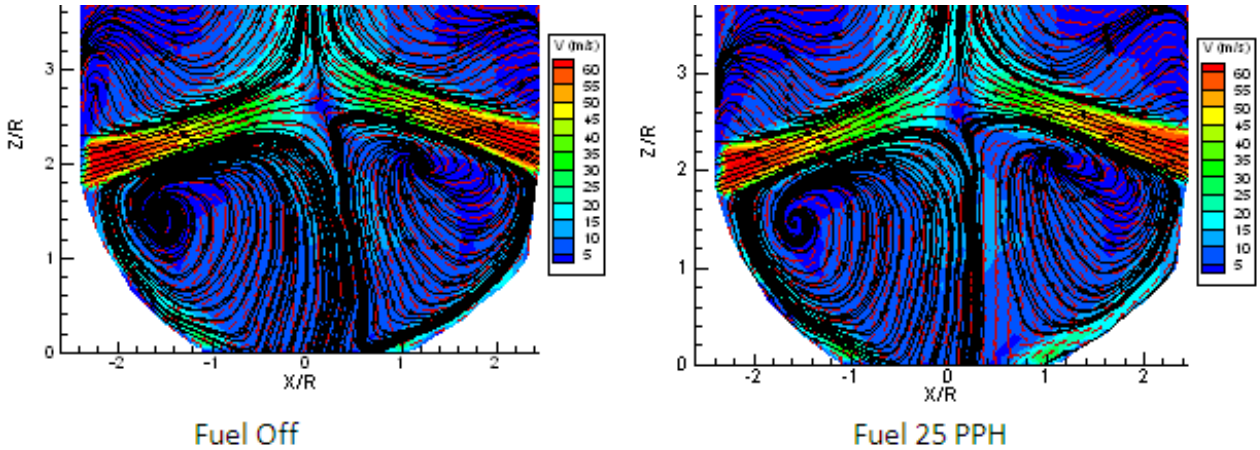

Figure 17. Velocity vectors and streamlines with and without fuel injection (fuel flow rate of 25 pphr).

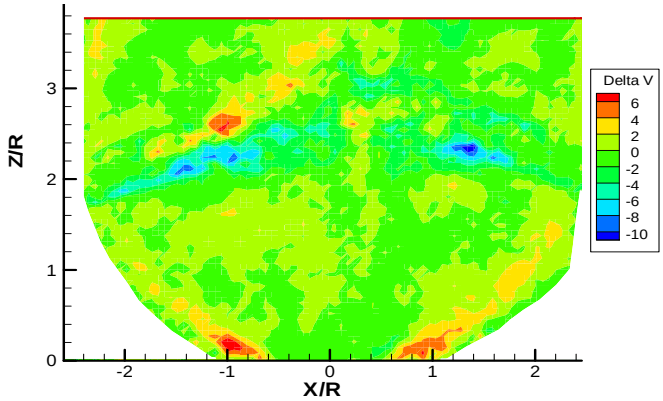

Figure 18. Difference in velocity magnitudes with and without fuel injection at fuel flow rate of 25 pphr.

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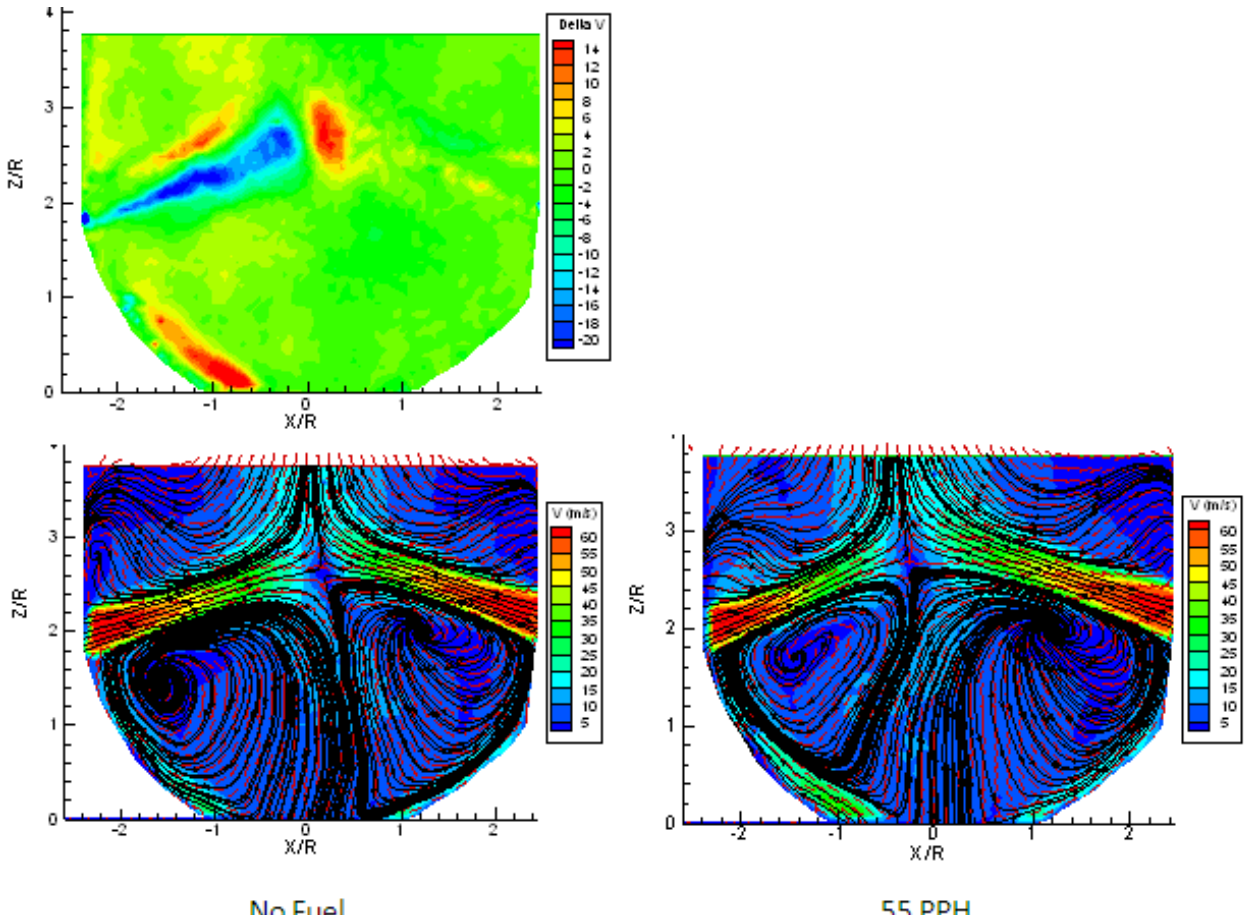

Figure 19. Fuel injection effect on the flow field structure at fuel flow rate of 55 pphr.

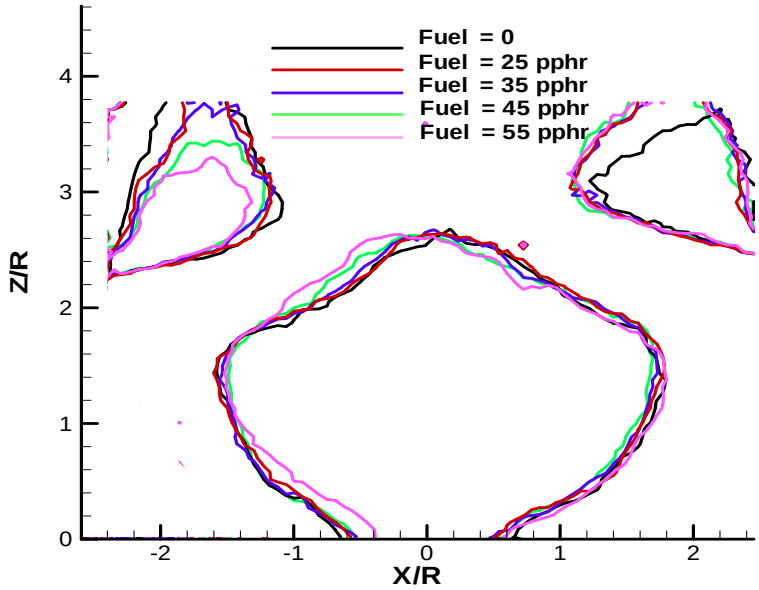

Figure 20. Effect of the fuel injection on the edges of the CRZ and Jets wake regions.

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